

2023 Ch H2 Q10

Section: Researching Chemistry

Topic: Gravimetric & Gas Collection
Analysis

Question Summary

- (a) Describe how to accurately weigh a 1.5 g mixture into a crucible.
- (b) (i) Explain why the lid is lifted during heating.
(ii) Explain why a Bunsen burner is a suitable heating method.
- (c) Use mass difference to calculate mass of MgCO_3 in the mixture.
- (d) (i) Draw and label suitable gas collection apparatus.
(ii) Explain why CO_2 can be collected over water.

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q10:

(a) The crucible is cleaned and dried. It is weighed empty, then the mixture is added. The balance is zeroed and the sample weighed accurately by difference until ~ 1.5 g is obtained.

(b)(i) The lid is lifted to allow carbon dioxide gas to escape, while preventing loss of solid by spitting.

(b)(ii) A Bunsen burner provides a continuous high temperature that is sufficient to fully decompose the carbonate.

(c) Mass loss = $1.598 - 1.294 = 0.304$ g
 CO_2 .

Moles $\text{CO}_2 = 0.304 \div 44.0 = 6.91 \times 10^{-3}$ mol.

Reaction ratio $\text{MgCO}_3 : \text{CO}_2 = 1 : 1 \rightarrow$ moles $\text{MgCO}_3 = 6.91 \times 10^{-3}$ mol.

Mass $\text{MgCO}_3 = 6.91 \times 10^{-3} \times 84.3 = 0.583$ g.

(d)(i) Apparatus: conical flask containing mixture + acid, delivery tube leading to an inverted measuring cylinder over a trough

of water.

(d)(ii) CO_2 is only slightly soluble in water, so it can be collected by displacement of water.

Final Answer

(a) Weigh accurately by difference into a crucible.

(b)(i) To allow CO_2 to escape, prevent spitting.

(b)(ii) Bunsen burner provides continuous high heat.

(c) Mass $\text{MgCO}_3 = 0.583 \text{ g}$.

(d)(i) Conical flask + acid → delivery tube → inverted measuring cylinder over water.

(d)(ii) CO_2 is only slightly soluble, so collect over water.

Revision Tips

- Always use weighing by difference for accuracy.
- Heating carbonates requires high heat until constant mass.
- Gravimetric analysis: mass loss → moles of gas → mass of reactant.
- Gas collection: use water displacement for gases not highly soluble in water.

